M661 Applied Statistics - Module 1 Assignment

Task 2: Titanic Data Analysis

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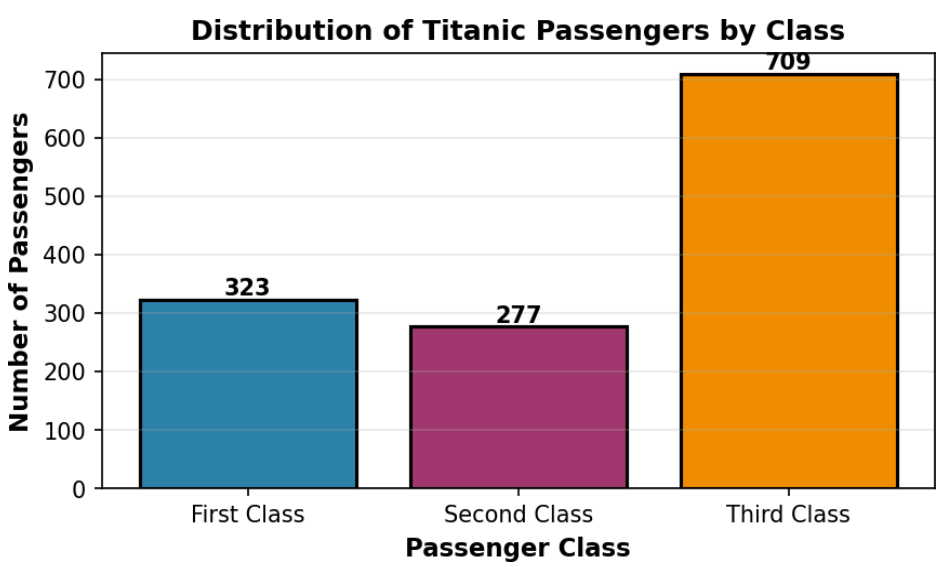
Background:

On April 15, 1912, on her maiden voyage, the Titanic collided with an iceberg and sank. The ship was luxurious but did not have enough lifeboats for the 2224 passengers and crew. As a result of the collision, 1502 people died. The ship had three classes of passengers. The level of luxury and the price of the ticket varied with the class, with first class being the most luxurious. There were 323 passengers in first class, 277 in second class, and 709 in third class.

Question 1 (15 points)

(a) Bar graph of passenger distribution by class

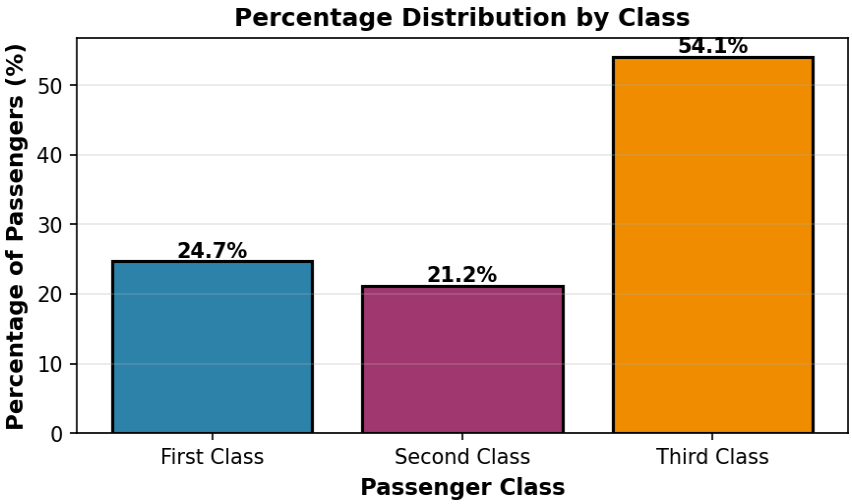
The bar graph below displays the distribution of Titanic passengers across the three classes. The x-axis represents the passenger class (First, Second, and Third), while the y-axis shows the number of passengers in each class. Different colors distinguish between classes, with the actual count displayed on top of each bar for clarity.



(b) Summary of passenger distribution

Summary: The bar graph reveals a notably unequal distribution of passengers across the three classes. Third class had the largest number with 709 passengers (54.1%), comprising more than half of all passengers. First class had 323 passengers (24.7%), representing about one-quarter of the total. Second class had the smallest number with 277 passengers (21.2%), just over one-fifth of the total. The distribution shows a clear inverse relationship between class luxury and passenger count: the higher the luxury level, the fewer passengers traveled in that class. Third class contained more than twice as many passengers as first class, reflecting the economic realities of early 20th-century transatlantic travel.

(c) Comparison: Count vs. Percentage Bar Graphs

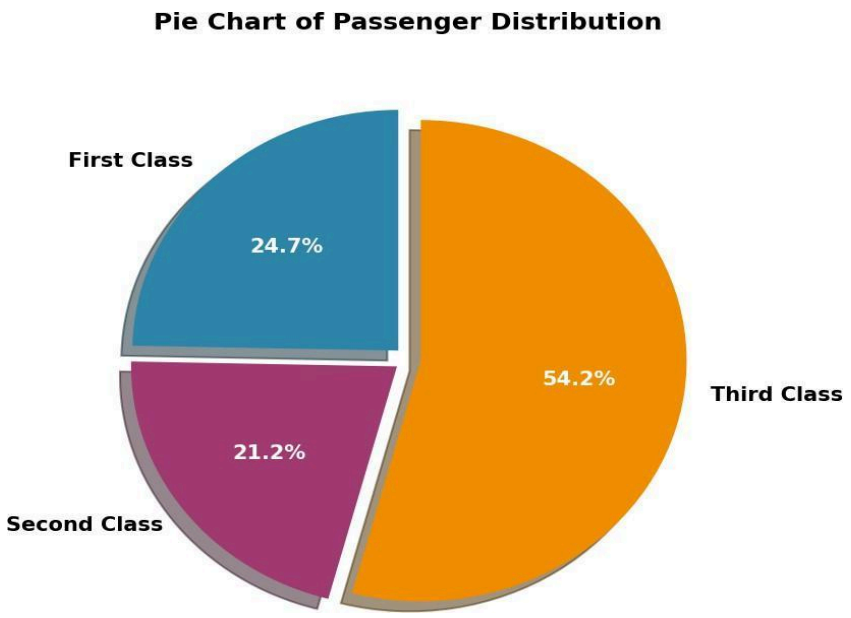


Analysis: If we created a bar graph displaying the *percentages* of passengers in each class instead of the raw counts, the **general features and pattern would remain the same**. The relative heights of the bars would be identical, the ordering (Third > First > Second) would be unchanged, and the visual pattern would look the same. What would change is only the y-axis scale (showing 0-100% instead of 0-800) and the numerical labels on the bars (54.1%, 24.7%, 21.2% instead of 709, 323, 277). This is because percentages are simply the counts divided by the total and multiplied by 100—a linear transformation that scales all values proportionally while preserving their relative relationships.

Question 2 (10 points)

(a) Pie chart of passenger distribution

The pie chart below displays the same passenger distribution data, with each slice representing one of the three passenger classes. The size of each slice is proportional to the percentage of passengers in that class. Each slice is labeled with both the class name and the percentage.



(b) Comparison: Pie Chart vs. Bar Graph

Bar Graph Advantages: (1) Easier to read exact values from the y-axis or labels, (2) Better for comparing differences—comparing bar heights is more intuitive than comparing angles, (3) Clearer ordering of categories is immediately apparent, (4) Can accommodate many categories without becoming cluttered, (5) Facilitates showing additional data dimensions through stacked or grouped bars.

Pie Chart Advantages: (1) Excels at showing part-to-whole relationships and emphasizes that all slices sum to 100%, (2) Visually intuitive for seeing that third class represents "about half" of passengers, (3) Compact presentation in a single circular format, (4) The complete circle reinforces totality.

My Preference: I prefer the **bar graph** for this dataset because: (1) It makes precise comparisons easier—we can clearly see that third class has 2.56 times as many passengers as second class, (2) The hierarchical nature of first/second/third class ordering is preserved naturally, (3) Exact values are more important here than approximate proportions for understanding the disaster's impact, (4) Bar graphs are more versatile for showing survival data later (Question 3), allowing stacked or grouped bars. However, if the goal were simply to communicate "about half were in third class," the pie chart would be more effective for that single message.

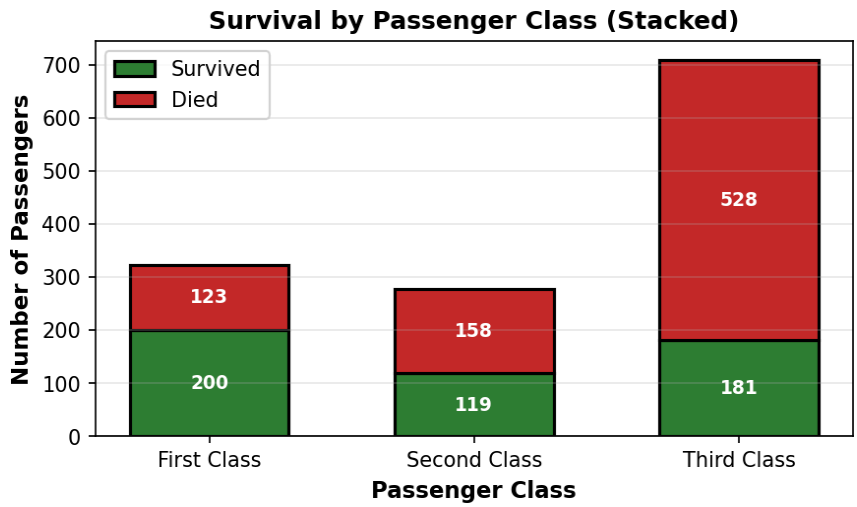
Question 3 (5 points)

Graphical summary of survival by passenger class

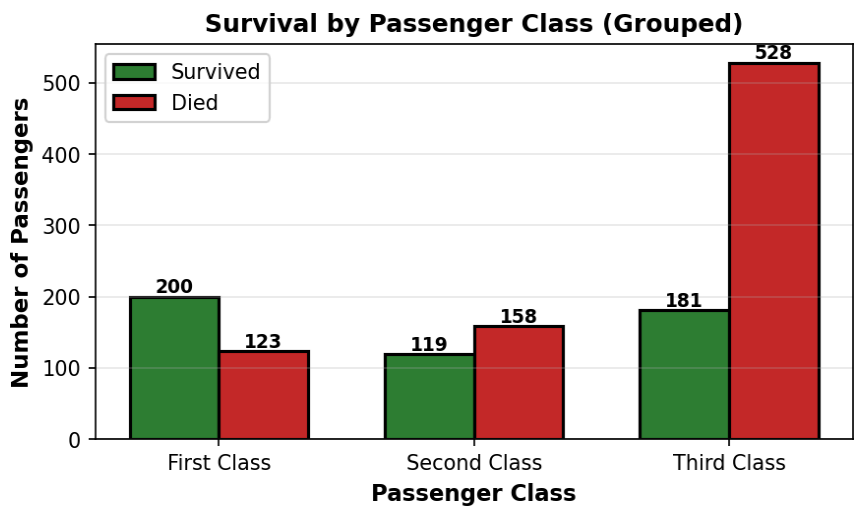
Survival Data: First Class: 200 survived out of 323 (61.9% survival rate); Second Class: 119 survived out of 277 (43.0% survival rate); Third Class: 181 survived out of 709 (25.5% survival rate).

I have created three complementary visualizations to show how survival depended on passenger class:

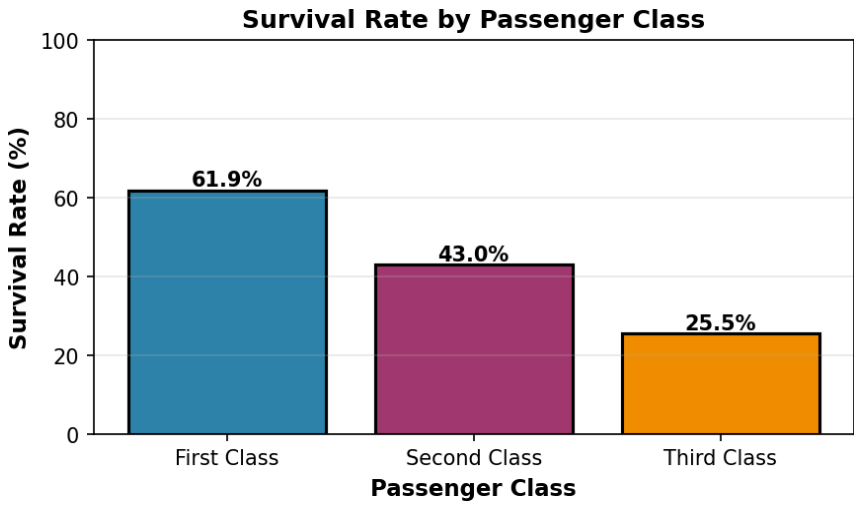
Stacked Bar Chart: Shows survived (green) and died (red) for each class.



Grouped Bar Chart: Places survived and died side-by-side for direct comparison.



Survival Rate Chart: Shows percentage who survived in each class.



Key Findings: These visualizations reveal a **strong relationship between passenger class and survival**. First class had the highest survival rate (61.9%)—nearly two-thirds survived. Second class had an intermediate rate (43.0%)—less than half survived. Third class had the lowest survival rate (25.5%)—only one in four survived, despite representing over half of all passengers. This resulted in 528 deaths in third class alone, accounting for more than 70% of all deaths. The data shows survival was inversely related to class: the higher the class (and presumably wealth/status), the better the chance of survival. This reflects both the ship's physical layout (first-class cabins closer to the boat deck) and potential social factors in the evacuation process.